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# Fluid reasoning, working memory and planning ability in assessment of risk for mathematical difficulties

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## ABSTRACT

The demands on mathematical problem-solving have increased in almost all school systems internationally and may constitute a barrier for children with special educational needs (SEN). This study explored the role of fluid reasoning (FR), working memory (WM) and complex executive function of planning (EF) in children ( $N = 62$ ) referred for assessment of SEN, and specifically of risk for mathematical difficulties (MD). Performances on FR, WM and complex EF of planning were used to predict risk for MD. Results showed that planning ability predicted children at risk for MD, beyond FR or WM ability, when comparing with children not at risk for MD. It was concluded that assessing the complex EF of planning in addition to FR and WM ability is crucial in identifying children at risk for MD. The importance of understanding how planning ability affects children's mathematical problem-solving is discussed, in relation to assessment and teaching practices.

## KEYWORDS

Psychological assessment; fluid reasoning; working memory; executive function; planning; mathematical learning difficulties

## Introduction

In recent decades there has been an increased focus on mathematical problem-solving and reasoning as part of curricular goals and instruction in almost all school systems internationally (Mullis et al., 2016). Problem-solving skills are included in vital parts of standards for mathematical education, such as Common Core State Standards in the US (Initiative, C. C. S. S. M., 2010), and national curriculums in European countries, for example, England and Sweden (Department for Education, 2013; Skolverket, 2016). Problem-solving skills are described in this context as students' abilities to organise information, make decisions, plan and apply strategies and solve non-routine problems. For students, this means that there is a trend towards higher levels of cognitive demand in mathematics. For teachers, understanding how these skills develop and how students can be supported for optimal learning will be important. This is particularly interesting considering that international assessments of mathematical knowledge (including problem-solving and advanced mathematics) among students show decreased results in many countries (Mullis et al., 2016).

The increased cognitive demands on problem-solving also have consequences for special education teachers and psychologists working with assessing children's special educational needs (SEN) and identification of children at risk for mathematical difficulties

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(MD). These children often struggle with working memory (WM) and executive functions (EF), which are considered as domain-general abilities affecting learning in a broad range of school-related subjects. For example, research has shown that difficulties with WM and EF contribute to learning difficulties among children identified as having SEN (Alloway et al., 2005) and that WM and EF particularly are underlying mathematical skill development in primary school children (Friso-van den Bos et al., 2013).

In an overview of the scientific debate regarding mathematical difficulties, Geary (2013) concluded that the early foundations for mathematics learning include learning a system of representing number magnitudes (quantities), as well as their relation to each other. Besides these two, a third system was stressed by Geary: the attention-control and problem-solving system that enables the processing of mathematical thinking.

With regards to WM, in the original Baddeley (2007) multicomponent model, three components were described: a visuospatial and a phonological component responsible for temporary storage of short-term information, and a central executive responsible for more elaborate processing of information. Following this tripartite division, a substantive amount of research has indicated that WM tasks predict performance in higher-order cognitive tasks, including maths tasks (Geary, 2011; Raghubar et al., 2010; Unsworth et al., 2009). Since mathematical performance involves a broad range of abilities, such as conceptual understanding of numbers, number estimation, counting concepts, arithmetics and problem-solving, it can be expected that broad domain-general cognitive abilities (that is, abilities affecting learning across several subjects) like WM are involved in the development of MD (Geary et al., 2007). On the other hand, some researchers have focused on the central executive processes and EF in learning. As suggested by Miyake et al. (2000), EF is concerned with the inhibition, shifting and updating of the mental content and WM, as a contrast to simple WM operations such as WM span and storage.

In this paper, the authors adhere to the division of EF proposed by some researchers, where *simple* EF is concerned with executive aspects of attentional processes of sustaining, shifting attention and inhibition (Lehman et al., 2010), while *complex* EF involves planning ability, which is related to goal-oriented thinking, strategy use, self-monitoring and metacognitive abilities involved in problem-solving. The complex EF of planning is, similarly to simple EF, a domain-general ability predicting academic achievement including maths (Best et al., 2011; Bull & Lee, 2014). This division in simple and complex EF might be of importance since others have advocated that EF serves as an important predictor of MD and mediates in the development of mathematical skills (Bull & Lee, 2014; Toll et al., 2011), findings that probably imply more complex involvement in such development. The international school trend, with increased focus on mathematical problem-solving and higher-order thinking as part of curriculum and instruction, also serves as a contextual background to why the complex EF of planning ability can be hypothesised to play an increasing role in mathematics learning in schools today.

However, understanding the roles of WM and complex EF of planning in mathematics also requires taking a general cognitive factor into consideration. It has been widely accepted that general cognitive ability (or IQ) plays an important role in school achievement, particularly with the demands on fluid reasoning in mathematics (Deary et al., 2007; Taub et al., 2008). Fluid reasoning has been defined as the ability to analyse problems, to identify patterns and relationships in information, and apply logic, particularly to novel

information. Fluid reasoning, as a broad cognitive ability, has been shown to be strongly related to mathematical performance in children and adolescents (Green et al., 2017).

A common practice in identification and remediation of children's SEN, including determining eligibility for children at risk for learning difficulties, has been to use the so-called "discrepancy model", in which a difference between general cognitive ability (intelligence) and school achievement (as measured by standardised tests or observed in school subjects) is used as a criterion for identification of children at risk for learning difficulties (Lebeer et al., 2012; Restori et al., 2009). However, the discrepancy model has received criticism since it has been shown that it does not reliably differentiate children at risk from children who are low achieving (Fletcher et al., 2001). Also, in the subject area of mathematics, studies of WM and EF have shown that children with simultaneous arithmetic and WM deficits are not reliably differentiated by a discrepancy between IQ/general cognitive ability and mathematics achievement, when comparing children with low and normal general cognitive ability (Maehler & Schuchardt, 2011). In recent years, more process-oriented and neuropsychological assessment models have developed and, rather than focusing on general cognitive ability, assessing both WM and to some extent EF has become an integral part of special educational, clinical and school psychological practices in many countries. It has also been advocated that assessment practices should be informed by research in the area of neuropsychology and related fields which can contribute with a deeper understanding of the specific cognitive demands in different areas of learning (Hale et al., 2006).

For special education teachers and school psychologists working with identification and remediation of children's SEN, assessing FR and WM abilities has been an integral part of the assessment of student's general cognitive abilities (that is, intelligence testing) with, for example, the Wechsler Intelligence Scales (Wechsler, 2003, 2014). However, explicit tasks of complex EF planning involved in problem-solving are missing in the Wechsler scale constructs, leaving practitioners with little information about the student's abilities in these areas. Therefore, other approaches to cognitive ability testing which include explicit EF measures have been developed, for example, the Das-Naglieri Cognitive Assessment System (CAS, Naglieri & Das, 1997; Naglieri et al., 2014) that includes both simple EF in the form of attention measures, and complex EF in form of planning measures (Best et al., 2011). The complex EF of planning is related to strategy use and metacognitive abilities that require the child to create a plan of action, apply that plan, and to monitor the effectiveness of the plan. Planning processes have thus been stressed as important and somewhat overlooked aspects that contribute particularly to the development of problem-solving abilities in mathematical skill development (Cai et al., 2016). Research has also shown that students with MD exhibit cognitive weaknesses on planning processes (Kroesbergen et al., 2003).

Given the trend towards problem-solving and higher-order thinking in mathematical education, the authors wanted to examine to what extent this was also reflected in the population of children with potential special educational needs (SEN), a group that is frequently referred for assessment to special education teachers and psychologists. The authors also wanted to explore to what extent WM and complex EF planning measures used in assessment could contribute to the understanding of children at risk for mathematical learning difficulties, beyond the role of fluid reasoning (FR). In the current study, the cognitive characteristics in referred children with potential SEN at risk and not at risk

for MD are explored. Also examined is the role of FR, WM and complex EF planning in predicting risk for MD as measured by the Perceptual Reasoning and Working Memory indices of WISC-IV and the Planning index of CAS. It was hypothesised that FR, WM and Planning would all contribute to the prediction of risk for MD, but that complex EF as measured by Planning would do this to a greater degree than both FR and WM, given the increased emphasis on mathematical problem-solving in school.

## Method

### *Participants*

Subjects ( $N = 62$ ) were children referred to school psychologists in Sweden for SEN assessment and remediation. Based on descriptions from teachers and/or special education teachers, psychologists made a classification of the specific area of difficulties that constituted the reason for referral, in this case at risk for mathematical difficulties (MD). The definition of “at risk for MD” was based on low achievement in maths at a level which puts the child at risk for not meeting the criteria in the maths curriculum goals, and according to the Swedish legislation for SEN support (Skollagen, 2010). The purpose of assessment at this stage is primarily to gain deeper knowledge of cognitive functions of the child, and to inform intensified instruction, corresponding to a tier 2 intervention (Hoover & Patton, 2008). Children with explicit intellectual disability as a reason for referral were excluded. Children with missing data from either WISC-IV or CAS assessments were also excluded. The final study included children ( $N = 62$ ) between ages of 6 and 16 years ( $M = 10.8$  years,  $SD = 2.5$ , 43 males and 22 females). Of these children, 38 were classified by the psychologists as at risk for MD and 24 not at risk.

### *Measures and procedure*

In the study, fluid reasoning (FR) was measured with tasks from the Perceptual Reasoning index, and working memory (WM) with tasks from the Working Memory index of the WISC-IV. The tasks in Perceptual Reasoning are typical tasks tapping fluid reasoning (Otero, 2017; Wechsler, 2003; Williams et al., 2003) while Working Memory tasks tap verbal, phonological and visuospatial WM processing (Crowe, 2000; Leffard et al., 2006). Furthermore, the Planning tasks from the Das-Naglieri Cognitive Assessment System (CAS) were used to measure complex EF (Naglieri & Das, 1997; Naglieri et al., 2014). The CAS was originally developed as part of a new conceptualisation of neurocognitive processing based on a four-block model of intellectual functioning: Planning, Attention, Simultaneous and Successive processes (PASS-theory), based on the neuropsychological work of Luria (1976). Moreover, the construct and predictive validity of both the WISC-IV (Wechsler, 2003) and the CAS (Naglieri & Rojahn, 2004) have been judged satisfactory and both are established standardised tests used by practitioners (Flanagan & Harrison, 2012).

Trained licenced psychologists administered the tests. The assessments took place during a period of maximally 6 months between the CAS and WISC-IV-tests and were conducted individually following standardised procedures and with parental consent given prior to assessment.

Standard scores of the FR and WM ( $M = 100$ ,  $SD = 15$ ) were drawn from the WISC-IV, using the British standardisation norms. The subtests Block Design, Picture Concepts, and Matrix Reasoning were included among FR tasks. In *Block Design* the child is asked to put together red and white blocks in a pattern according to a displayed model, with increasing complexity. The tasks are timed and the child is allowed to work until a specified level of omissions. The *Picture Concepts* subtest requires the child to recognise and categorise each of two or three rows of objects, and select objects that go together based on an underlying concept. In the *Matrix Reasoning* subtest, the child is presented with tasks containing a partially filled grid and asked to select the item that properly completes the matrix, from a range of options.

The WM subtests included were Digit Span, Letter-Number Sequencing and Arithmetics tapping short-term and WM span and mental manipulation (Leffard et al., 2006; Wechsler, 2003). The *Digit Span* subtest consists of a series of tasks where the child is required to listen and repeat a sequence of digits, with increasing level of difficulty. The task is administered forwards and backwards. The *Letter-Number Sequencing* subtest involves listening to and remembering a string of digits and letters read aloud at a speed of one per second. The child is then asked to recall the information by repeating the numbers in chronological order, followed by the letters in alphabetical order. The *Arithmetics* subtest consists of arithmetic problems of increasing difficulty administered orally. The child is asked to solve the task mentally without relying on paper and pencil.

A standard score of the Planning index ( $M = 100$ ,  $SD = 15$ ) was drawn similarly from the CAS, using norms from the US standardisation (Naglieri & Das, 1997). The Planning index subtests consist of three complex EF subtests: Matching Numbers, Planned Codes and Planned Connections, which present the child with novel tasks that require planning, monitoring and strategic problem-solving behaviour. The *Matching Numbers* subtest consists of four pages, each with eight rows of numbers, six numbers per row. The task is to find and underline the two numbers that are the same in each row. The numbers increase in length from one digit to seven digits. The *Planned Codes* subtest consists of two pages, each with distinct sets of codes and arrangements of rows and columns. A legend at the top of each page shows how letters correspond to simple codes (for example, A, B, C, D correspond to OX, XX, OO, XO, respectively). Each page contains seven rows and eight columns of letters without codes. The task is to translate letters into specific codes and fill in the corresponding codes in the pages. Children are permitted to complete each page in whatever fashion they desire. The *Planned Connections* subtest contains eight pages with boxes containing numbers or letters. The task is to connect the boxes, drawing a line with a pen. The first six items require children to connect boxes with numbers in a sequence (for example, 1–2–3–4–5). The last two items require children to connect the boxes with either numbers or letters in sequential order, alternating between numbers and letters (for example, 1-A-2-B-3-C). All three Planning subtests have time limits.

The Perceptual Reasoning, Working Memory and Planning index composite scores were chosen for analysis since research, as well as clinical recommendations for psychological assessment, suggests that interpretation should not rely on subtest scores but on composite index or full-scale scores (Naglieri & Das, 1997; Prifitera et al., 2008; Watkins, 2003). The Perceptual Reasoning and Working Memory indices of the WISC-IV, as well as the Planning index of CAS, have shown good test–retest reliability and the subtests have

shown satisfactory internal consistency respectively within the indices (Naglieri & Das, 1997; Wechsler, 2003).

## Statistical analyses

Descriptive statistics are presented in Table 1. To determine if the mean scores differed significantly between the MD and non-MD group, independent t-tests were conducted on FR, WM and Planning measures. Effect size was calculated with Cohen's  $d$ , defined as the difference between the means for individual groups, as a proportion of the pooled standard deviation. Cohen (1988) defined the magnitudes of effect sizes as follows: small,  $d = 0.2$ , medium,  $d = 0.5$  and large,  $d = 0.8$ . Furthermore, a correlational analysis (Pearson Correlation, two-tailed) was conducted in order to examine the association between the FR, WM and the Planning index. Finally, a logistic regression analysis was performed to examine the unique contribution of FR, WM and Planning to risk for MD. In Step 1 the authors examined the effect of FR in predicting risk for MD, with WM entered in Step 2 and Planning entered in Step 3. Then, a second logistic regression was examined in order to examine the unique contribution of Planning to prediction, and also the relation between Planning and WM, by entering Planning in Step 1 and entering WM in Step 2.

## Results

As can be seen in Table 1, the SEN children's performances on FR, WM and Planning were significantly lower than the standardised population mean, and for WM and Planning also substantially below the cut-off criterion (81 standard score) for low performance widely used in research (Fletcher et al., 1998; Mayes et al., 1998). When SEN children at risk and not at risk for MD were compared, FR, WM and Planning all significantly differed between groups (with a medium magnitude of effects for FR and WM, and large magnitude of effects for Planning). The correlational analyses showed that there was a significant moderate positive correlation between all measures. The correlation between FR and WM was  $r(60) = .324$ ,  $p = .010$ , between FR and Planning was  $r(60) = .393$ ,  $p = .002$  and between WM and Planning was  $r(60) = .388$ ,  $p = .002$ .

Table 2 presents the results from the two logistic regression analyses. When FR was entered in Step 1, the model was statistically significant in predicting risk for MD,  $\chi^2(1) = 3.962$ ,  $p = .047$  and explained 8% of the variance (Nagelkerke  $R^2$ ). When WM was entered in Step 2, no significant improvement in model fit occurred,  $\chi^2(1) = 2.028$ ,  $p = .154$ . When Planning was entered in Step 3, a significant improvement in model fit occurred,  $\chi^2(1) = 7.500$ ,  $p = .006$  and the explained variance increased to 27%. When examining the

**Table 1.** Descriptive statistics for total sample of SEN children.

|                 | Total sample ( $n = 62$ ) |       | MD ( $n = 24$ ) |       | non-MD ( $n = 38$ ) |       | non-MD vs MD |     |
|-----------------|---------------------------|-------|-----------------|-------|---------------------|-------|--------------|-----|
|                 | $M$                       | $SD$  | $M$             | $SD$  | $M$                 | $SD$  | $p$          | $d$ |
| Fluid Reasoning | 87.19                     | 14.20 | 84.39           | 14.20 | 91.63               | 13.30 | .050         | .53 |
| Working Memory  | 74.34                     | 13.50 | 71.55           | 13.40 | 78.75               | 12.69 | .040         | .55 |
| Planning        | 75.48                     | 12.29 | 71.37           | 8.53  | 82.00               | 14.54 | .003         | .89 |

All measures presented as standard scores ( $M = 100$ ,  $SD = 15$ ).



**Table 2.** Two logistic regressions predicting likelihood of risk for MD.

| Step           | Variable | <i>B</i> | <i>SE</i> | Wald | <i>df</i> | <i>p</i> | Odds Ratio | 95% CI for Odds Ratio |       |
|----------------|----------|----------|-----------|------|-----------|----------|------------|-----------------------|-------|
|                |          |          |           |      |           |          |            | Lower                 | Upper |
| 1 <sup>a</sup> | FR       | −.038    | .02       | 3.66 | 1         | .056     | .96        | .93                   | 1.00  |
| 2              | FR       | −.027    | .02       | 1.60 | 1         | .206     | .97        | .93                   | 1.01  |
|                | WM       | −.031    | .02       | 1.95 | 1         | .162     | .97        | .93                   | 1.01  |
| 3              | FR       | −.011    | .02       | .246 | 1         | .620     | .99        | .95                   | 1.03  |
|                | WM       | −.022    | .02       | .832 | 1         | .362     | .98        | .93                   | 1.03  |
|                | Planning | −.074    | .03       | 6.21 | 1         | .013     | .93        | .88                   | .98   |
| 1 <sup>b</sup> | Planning | −.085    | .03       | 8.91 | 1         | .003     | .92        | .87                   | .97   |
| 2              | Planning | −.077    | .03       | 7.07 | 1         | .008     | .93        | .88                   | .98   |
|                | WM       | −.026    | .02       | 1.24 | 1         | .265     | .98        | .93                   | 1.02  |

<sup>a</sup>FR entered in Step 1, WM entered in Step 2, Planning entered in Step 3.

<sup>b</sup>Planning entered in Step 1, WM entered in Step 2.

variables in the final model, only Planning contributed significantly to explain at risk for MD, while FR and WM failed to contribute significantly.

In the second logistic regression, when Planning was entered in Step 1, the model was statistically significant,  $\chi^2(1) = 11.974$ ,  $p = .001$ , explaining 21% of the variance (Nagelkerke  $R^2$ ). When WM was entered in Step 2, no significant improvement in model fit occurred,  $\chi^2(1) = .270$ ,  $p = .260$ . The final model was significant,  $\chi^2(2) = 13.244$ ,  $p = .001$ , and WM did not contribute significantly to the model, leaving Planning as the only significant predictor.

## Discussion

The purpose of the present study was to explore to what extent WM and complex EF measures used in SEN assessment could contribute to the prediction of risk for MD, beyond the role of fluid reasoning (FR). The authors also wanted to examine the cognitive characteristics of children with SEN at risk and not at risk for MD. Given the shift in mathematical education towards more complex, higher-order thinking and problem-solving, the authors predicted that these increased cognitive demands should be reflected in a typical referred population of children with SEN, and particularly increasing the predictive role of complex EF (as measured by Planning).

The results showed that the referred group of SEN children scored somewhat low on FR and particularly low on measures of both WM and Planning, suggesting that WM and complex EF are cognitive functions challenged in children with SEN. This is in line with previous studies showing that low WM as well as problems with EF are common among children with SEN (Alloway et al., 2005; Partanen et al., 2015; Pickering & Gathercole, 2004).

However, comparing SEN children at risk and not at risk for MD showed that while FR and WM differed significantly between groups with moderate magnitude effects, complex EF (Planning) did this to a large magnitude of effect. This finding is in line with previous research (Bull & Lee, 2014; Kroesbergen et al., 2003; Toll et al., 2011). The results also showed that FR, WM and complex EF are correlated constructs, which has earlier been observed in studies concerned with the relationship between fluid reasoning abilities and EFs (Decker et al., 2007), as well as between simple WM and more complex EFs (Unsworth et al., 2009).

Further, the logistic regression analyses showed that only complex EF (as measured by Planning) predicted risk for MD in the referred group of SEN children, over and above FR



and WM. The analyses also showed that complex EF alone explained 24% of the variance. The relatively modest significant contribution of FR, with a small amount of explained variance (8%), and the non-significant contribution of WM was a surprising outcome considering research on the general role of WM (Unsworth et al., 2009), but was in line with research which more specifically addresses the role of EFs in learning, maths achievement and MD (Best et al., 2011; Bull & Lee, 2014; Toll et al., 2011). Even though all the significant effects of Planning fell well below the conventional  $p < .05$ , it is important to recognise that the presence of more than one explanatory variable in the regression analyses, separately testing each independent variable against the response variable inflates the alpha, and thus increases Type I error rate.

### ***Conclusion, limitations and implications for practice***

Several limitations of the present study are worth noting. The study has a small sample and, given results from previous research, it cannot be ruled out that WM adds to the predictive power of assessing the risk for MD. Another limitation was that the definition of “at risk for MD” was based on the evaluations of psychologists and special education teachers, and should not be confused with an actual diagnosis of mathematical learning disabilities. As such, the definition of “at risk for MD” does not inform in detail which mathematical tasks or skills are challenged. However, this limitation is not only restricted to the current study, as this is also the case in the formal diagnosis of mathematical learning difficulties, which has been studied using a variety of different mathematical achievement measures and diagnostic criteria (Geary et al., 2007). Additionally, the authors did not use mathematical achievement scores in defining risk for MD, which could have increased explained variance if a continuous outcome variable had been used. However, observations and assessments by mainstream teachers, special education teachers, and school psychologists of children and their functioning in learning go beyond mathematical achievement scores, and it could be argued that relying on their assessments increases the ecological validity in this referred population of children with SEN. Thus, further research could address the above-mentioned limitations.

It has been acknowledged that studying the role of WM and EF in learning is complicated since these constructs are intertwined with numerous definitions and models (Fenesi et al., 2014; Hofmann et al., 2011; Unsworth et al., 2009), and particularly in conjunction with complex environments, such as in school, and in maths education. However, despite these limitations, the results suggest that complex EF planning processes, in contrast to both fluid reasoning and WM, differentiate between at risk and not at risk for MD significantly and with reasonably large magnitude of effect, among referred children with SEN.

Although outside the scope of this study, it could be argued that the results also suggest that, following assessment of children with SEN at risk for MD, remediation should address training of complex EF planning processes in addition, or even instead of, training of WM. Since reviews of WM training (Melby-Lervåg & Hulme, 2012) have shown that WM training effects tend to be short-termed and transference to school subjects is rare, addressing more complex EF by training planning and metacognitive strategies could be a future direction, particularly in mathematical problem-solving. This has also been explored in a recent study, where complex EF was addressed by adding components of planning and metacognitive strategy training into WM training in school

(Partanen et al., 2015). In a review of WM and EF training, it has also been proposed that there is a need for more process-based training regimens (Titz & Karbach, 2014). Previous studies have also addressed the particular role of complex EF planning processes in improving mathematical abilities among children with learning difficulties (Iseman & Naglieri, 2011; Naglieri & Johnson, 2000).

The results from those studies suggest that the early and original definitions of executive functions (Das, 1980; Lezak, 1982; Luria, 1976), stressing the capacities of formulating goals, organising behaviour, planning, carrying out plans, and self-monitoring, are important dimensions in assessment and an integral part of the broad EF concept beyond definitions of WM. Later research on the division of simple and complex EF (Best et al., 2011; Lehman et al., 2010; Puhan et al., 2005) has further added to the understanding of the role of EF in learning and development, supporting the validity of planning ability as a complex EF. The results from the current study also support that planning and problem-solving abilities are important in mathematics today, particularly when considering the decreased results in mathematical knowledge in many countries, and the trend towards advanced mathematics and problem-solving in many countries' curricula (Mullis et al., 2016).

In conclusion, the results of the current study have consequences for the assessment practices of teachers, special educators, clinical and school psychologists, suggesting it is important to tap complex EF and planning ability when conducting general cognitive assessment at an early stage in referrals of children with SEN, especially when screening for risk for MD is desirable. Given the proposed role of deficit planning processes for children at risk for MD in the present study, it is also important to highlight how planning interacts with increased cognitive demands following mathematical problem-solving in school, via the school curriculum as well as in everyday practice. These demands in turn, if not understood, might constitute a barrier for optimal learning, and affect children's mathematical development detrimentally. This calls for increased attention from both teachers, special education teachers and school psychologists in early identification and prevention of difficulties in planning and problem-solving, and in promoting the development of these abilities among children in general, and children with SEN and at risk for MD in particular.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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